

Shashank Srinivasan

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Seattle, WA | Authorized to work in the US (STEM OPT, 36 months); no immediate sponsorship required

Senior Machine Learning Engineer working across HPC and ML systems: GPU and parallel computing, analog compute-in-memory, and large-scale data pipelines. Cut a pipeline over billions of events from roughly 5 hours to about 20 minutes, and researches drift correction for analog ML accelerators. Holds an MS in CS (HPC specialization) from Georgia Tech, May 2026.

EXPERIENCE

Senior Machine Learning Engineer, Walmart | *Dec 2024 – Jul 2025*

- Cut per-experiment compute cost by **40%** on floor-price and adtech recommendation ML pipelines by profiling PySpark bottlenecks, rewriting shuffle-heavy transformations, and right-sizing batch processing.
- Built a **Text2SQL agentic system** with plain-English querying, visualizations, and active chat, giving non-technical teams self-serve analytics over business and ML pipeline data.
- Improved production ML reliability and release velocity by building an **observability framework** and standardizing model build, test, and deployment with GitHub Actions.

Machine Learning Engineer III, Glance (InMobi) | *Jul 2022 – Dec 2024*

- **Founding engineer** on the rebuilt Glance feed personalization platform serving **100M DAU**; owned cold-start and sparse-user personalization end to end.
- Designed the sparse-user modeling stack: multinomial Thompson-sampling bandits, an **Elo-style exploration** scheme adapted from game matchmaking, and **lookalike models** (SVD/K-Means, then node2vec on app-ownership signals) that beat production baselines on time spent and reward; **+18.38%** time spent for cold users.
- Delivered content at **sub-50ms p99** via low-latency serving and streaming pipelines (prediction services, Vertex AI feature store, Kafka, ELK logging, Airflow retraining); **+12% conversion** and **+5% engagement**.
- Cut a rewards pipeline over **billions of clicks** from roughly 5 hours to about 20 minutes; standardized deployment, on-call, and automated canary across the team, mentored engineers, and led an org-wide paper-reading session.

Data Scientist II, Glance (InMobi) | *May 2021 – Jun 2022*

- Built Glance's **first live-commerce recommendation system** (Roposo) on graph embeddings at sub-50ms, and the feed team's first Python prediction service (sub-100ms, in two weeks): **+6.54%** lift over control and the best-performing model in production.
- Shipped real-time, NLP-based **quiz generation over live video** (GCP), demoed at the company AI Demo Day; integrated OEM and OTT content with external partners (Docubay, EpicON, Zee5).

Founding Data Scientist, Bright Money | *Jun 2019 – Apr 2021*

- Founded the data science function and built **Debt Manager**, Bright's flagship AI product that analyzes a user's finances and autonomously schedules debt payments; on Airflow and Django, it served **20K+ users/day** at **\$100K+ daily transaction volume** and drove close to **100% of revenue** by 2020.
- Shipped multiple 0-to-1 ML products: bill identification, affordability scoring, overdraft prediction, and liability inference for risk and retention workflows.
- Built time-series forecasting for income, expense, and account balance (Prophet, SARIMA) and segmentation models powering activation, retention, and monetization.

EDUCATION

Master of Science, Computer Science, Georgia Institute of Technology | *Aug 2024 – May 2026*

- Specialization: High Performance Computing.
- Research: Analog compute-in-memory error mitigation with Prof. Hyesoon Kim
- Coursework: GPU Hardware & Software, High-Performance Parallel Computing, HPC Architecture, Scientific ML, Deep Learning, Designing for Curiosity.

Bachelor of Engineering, Computer Science, Sir M. Visvesvaraya Institute of Technology | *May 2018*

- First Class with Distinction.
- Thesis: Image Regeneration with Generative Models; used CapsNet as a GAN discriminator for semantic inpainting on MNIST and face images.

SELECTED SYSTEMS & RESEARCH

Embedded Drift Sentinel (under review at ASPLOS 2027): Improved day-360 ImageNet **top-1 accuracy by +14 pp** on average over GDC across ResNet-18/34/50 and GoogLeNet on a 360-day PCM drift horizon (ResNet-50: 68% vs. 53% GDC, vs. 17% uncorrected). Designed a companion-crossbar architecture for online drift correction in analog compute-in-memory accelerators; eliminates main-array probe downtime at 0.3–1.7% cell overhead.

Physics-Informed RL for Plasma Control: Trained PPO policies in TORAX (auto-differentiable 1D core transport simulator) on a 50-parameter partially-observed state for ITER tokamak fusion control; achieved **1.7 ms average inference** (3.5 ms max), well under the 50 ms real-time threshold.

Meibomian Gland Segmentation: Published CNN-based meibography segmentation work in Biomedical Signal Processing and Control; developed custom augmentation and gland-health metrics validated against clinical expert annotations across tabletop and prototype handheld imagers.

Automated Debt Management Patents: R&D contributor; US20220261886A1, US20230075411A1.

HONORS & TEACHING

ACM ICPC 2015: Qualified to regional level; honorable mention.

Section Leader, Code in Place, Stanford University: Taught Python to 30+ students per cohort in 2021, 2023, and 2024.

TECHNICAL SKILLS

Languages: Python, C/C++, Go, SQL, PySpark, Shell

Artificial Intelligence & Machine Learning: Recommendation Systems, Candidate Generation, Retrieval & Ranking, Multi-Armed Bandits, Collaborative Filtering, Graph ML (node2vec, embeddings), Deep Learning, NLP, LLMs & Agentic Workflows (LangChain), Reinforcement Learning, Time-Series Forecasting (Prophet, SARIMA), A/B Testing; PyTorch, scikit-learn, XGBoost/LightGBM

MLOps & Infrastructure: Docker, Kubernetes, Airflow, Argo, Kafka, Spark, Vertex AI Feature Store, Neptune, GitHub Actions (CI/CD), ELK, Prometheus, Grafana; CUDA, MPI, OpenMP, GPU Profiling (Nsight Systems, nvprof), Distributed Systems, Scientific ML (PINNs)

Cloud & Data: GCP, AWS, Azure, PostgreSQL, Cosmos DB, Firebase, Tableau, Metabase, Django, Linux, Git